

Geometric-Mean Bridges Across Scale Hierarchies: From Planck to MOND

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Novelty, Significance, and Value

Context and prior work. The idea that geometric means bridge scale hierarchies in physics is not new. The Planck length as $GM(\text{Compton wavelength, Schwarzschild radius})$ is textbook.

Padmanabhan (2004) showed that the observed dark energy density satisfies $\rho_{\text{vac}} \approx \sqrt{(\rho_{\text{UV}} \cdot \rho_{\text{IR}})}$, the geometric mean of Planck-scale and Hubble-scale energy densities, and argued this reflects vacuum fluctuations rather than vacuum energy itself. Milgrom (2020) drew analogies between a_0 and other "boundary constants" relating seemingly independent scales. Capozziello et al. (2011) derived a_0 from scaling relations connecting Planck to cosmological scales. The geometric mean as a hierarchy-bridging tool is established in cosmology.

What is new. This paper extends the Padmanabhan program from the cosmological constant to the MOND acceleration scale. Papers 1-5 in this series (Nguyen 2026) established that $a_0 = cH_0(\Omega_{\Lambda}/\Omega_m)^{1/6}/(2\pi)$, which can be written as $a_0 = GM(c^2/C_{\text{now}}, c^2/C_{\text{eq}})$ — the geometric mean of horizon-scale accelerations at the current epoch and at matter-radiation equality. The present paper makes two specific contributions: (1) a classification of geometric-mean bridges into exact (running parameter cancels entirely) and approximate (running parameter is suppressed but survives), and (2) the identification of a_0 as an *approximate* bridge, formally distinct from the exact bridges at Planck and Compton scales. The $1/6$ -power residual dependence on $\Omega_{\Lambda}/\Omega_m$ is not a defect — it is the diagnostic signature of an approximate bridge, encoding the degree of asymmetry between the two epoch-dependent accelerations.

Why it matters. The "MOND coincidence" $a_0 \approx cH_0$ has persisted for 40 years without a structural explanation beyond numerology. Padmanabhan's geometric-mean observation addressed Λ but not a_0 . This paper closes that gap: a_0 is the geometric-mean fixed point of two cosmological accelerations, and the $\sim 15\%$ deviation from $cH_0/(2\pi)$ is the expected residual of imperfect parameter cancellation — the same mechanism that produces l_P , but with approximate rather than exact symmetry between the competing scales.

1. Introduction

Geometric means appear naturally at the boundaries between physical regimes. The Planck length $l_P = \sqrt{(\hbar G/c^3)}$ is the geometric mean of the Compton wavelength and Schwarzschild radius for any mass — a standard result in quantum gravity. The Compton wavelength $\lambda_c =$

$\hbar/(m_e c)$ is the geometric mean of the Bohr radius and classical electron radius — a standard result in atomic physics.

Padmanabhan (2004) extended this pattern to cosmology, showing that the observed dark energy density is well approximated by the geometric mean of the Planck-scale (UV) and Hubble-scale (IR) energy densities: $\rho_{\text{vac}} \approx \sqrt{(\rho_{\text{UV}} \cdot \rho_{\text{IR}})}$. This observation reframed the cosmological constant problem as a fluctuation effect rather than a fine-tuning puzzle.

The MOND acceleration $a_0 \approx 1.2 \times 10^{-10} \text{ m/s}^2$ presents a parallel puzzle. Since Milgrom (1983), the numerical proximity $a_0 \approx cH_0$ has been noted without structural explanation. Papers 1-5 in this series (Nguyen 2026) derived a precise formula — $a_0 = cH_0 (\Omega_\Lambda/\Omega_m)^{1/6}/(2\pi)$ — and showed it can be written as the geometric mean of two cosmological accelerations at different epochs.

The present paper asks: is the MOND geometric mean the same structural pattern as the Planck geometric mean and the Padmanabhan geometric mean? We argue that it is, but with a key distinction. The Planck and Compton cases are *exact* bridges — the running parameter cancels entirely. The MOND case is an *approximate* bridge — the running parameter (the density partition Ω_Λ/Ω_m) is suppressed from a $\sim 1/2$ power to a $1/6$ power, but not eliminated. This exact/approximate classification, and its application to a_0 , is the new contribution of this paper.

2. The Geometric-Mean Bridge Theorem

2.1 Setup

Consider two physical scales S_A and S_B (lengths, accelerations, energies, or any dimensionally consistent quantities) that depend on a dimensionless running parameter p :

$$S_A(p) = S_0 \cdot p^{+\alpha}, \quad S_B(p) = S_0 \cdot p^{-\beta}$$

where S_0 is a base scale independent of p , and $\alpha, \beta > 0$. Scale A grows with p ; Scale B shrinks with p . Together they define a hierarchy whose logarithmic width is $(\alpha + \beta) \log p$.

2.2 The Bridge

The geometric mean of the two scales is:

$$\mathcal{B}(p) \equiv \text{GM}(S_A, S_B) = \sqrt{S_A \cdot S_B} = S_0 \cdot p^{(\alpha-\beta)/2}$$

2.3 Classification

Case I — Exact bridge ($\alpha = \beta$). The running parameter cancels entirely:

$$\mathcal{B} = S_0 = \text{const}$$

The bridge is a universal constant, independent of p for all values. It sits at the geometric center of the hierarchy. The fixed point p^* where $S_A(p^*) = S_B(p^*)$ satisfies $S_A(p^*) = S_B(p^*) = S_0$, and the bridge always returns this fixed-point value regardless of where in the hierarchy it is evaluated.

Case II — Approximate bridge ($\alpha \neq \beta$, $|\alpha - \beta| \ll \alpha + \beta$). A weak residual survives:

$$\mathcal{B}(p) = S_0 \cdot p^{\epsilon/2}, \quad \epsilon \equiv \alpha - \beta$$

When $|\epsilon| \ll \alpha + \beta$, the bridge suppresses the parameter dependence from order-unity exponents down to $|\epsilon|/2$. The result is a "soft constant" — nearly parameter-independent, drifting slowly with p .

2.4 Why geometric means

The arithmetic mean $(S_A + S_B)/2 \approx S_A/2$ when $S_A \gg S_B$. It is dominated by the larger scale and provides no bridge. The geometric mean weights both scales equally on a logarithmic basis:

$$\log \mathcal{B} = \frac{1}{2}(\log S_A + \log S_B)$$

This is the natural interpolation when the hierarchy spans many orders of magnitude. More precisely, the geometric mean is the unique mean that maps multiplicative structure to additive structure, making it the natural midpoint between power-law scalings.

2.5 The bridge as a fixed-point detector

At the fixed point $p = p^*$ where $S_A(p^*) = S_B(p^*)$, the geometric mean equals both scales trivially. For an exact bridge, the remarkable property is that this equality holds for *all* p , not just p^* . The bridge "remembers" the crossing point everywhere. For an approximate bridge, the value drifts slowly away from the fixed-point value but remains close — it acts as a soft attractor in the space of scales.

3. Known Exact Bridges

Before presenting the new result, we note two well-known exact bridges (Case I) that establish the pattern.

The Planck length. For any mass m , the reduced Compton wavelength $\ell_{\text{quantum}} = \hbar/(mc)$ scales as m^{-1} and the gravitational radius $\ell_{\text{grav}} = Gm/c^2$ scales as m^{+1} . Their geometric mean is

$GM(\ell_{\text{quantum}}, \ell_{\text{grav}}) = \sqrt{(\hbar G/c^3)} = l_P$. The mass cancels exactly ($\alpha = \beta = 1$), yielding a universal constant. The Planck mass $m_P = \sqrt{(\hbar c/G)}$ is the fixed point where $\ell_{\text{quantum}} = \ell_{\text{grav}}$.

The Compton wavelength. The Bohr radius $a_B = \lambda_e/\alpha$ scales as α^{-1} and the classical electron radius $r_e = \lambda_e \cdot \alpha$ scales as α^{+1} , where $\alpha \approx 1/137$ is the fine-structure constant. Their geometric mean is $GM(a_B, r_e) = \lambda_e = \hbar/(m_e c)$. The electromagnetic coupling cancels exactly ($\alpha = \beta = 1$), yielding the pure quantum-relativistic scale of the electron.

Both are standard results. The question is whether this pattern extends beyond exact bridges to approximate ones — and in particular, whether the MOND acceleration scale a_0 is an instance of the same mechanism.

4. The MOND Acceleration Scale (Approximate Bridge)

4.1 The result from Papers 1–5

Papers 1–5 in this series (Nguyen 2026) established that the MOND acceleration scale can be written as:

$$a_0 = \frac{cH_0}{2\pi} \left(\frac{\Omega_\Lambda}{\Omega_m} \right)^{1/6}$$

and that this has a geometric-mean form:

$$a_0 = GM\left(\frac{c^2}{\mathcal{C}_{\text{now}}}, \frac{c^2}{\mathcal{C}_{\text{eq}}}\right)$$

where $c^2/\mathcal{C}_{\text{now}}$ and $c^2/\mathcal{C}_{\text{eq}}$ are horizon-scale accelerations at the current (Λ -dominated) epoch and at matter-radiation equality, respectively.

4.2 Identifying the running parameter

Define the base acceleration scale:

$$a_H \equiv \frac{cH_0}{2\pi}$$

and the running parameter:

$$R \equiv \frac{\Omega_\Lambda}{\Omega_m} \approx 2.3$$

The two cosmological accelerations can be written as:

$$a_{\text{now}} = a_H \cdot f_{\text{now}}(R), \quad a_{\text{eq}} = a_H \cdot f_{\text{eq}}(R)$$

The current-epoch acceleration is dominated by the de Sitter horizon, giving $a_{\text{now}} \propto \Omega_{\Lambda}^{1/2}$, which at fixed total density budget scales as $R^{1/2}$ to leading order. The equality-epoch acceleration depends on Ω_m and Ω_r ; since Ω_r is fixed by the CMB temperature and the equality redshift $z_{\text{eq}} \propto \Omega_m/\Omega_r$, its R -dependence enters only through the $\Omega_m = 1/(1+R)$ constraint.

The key observation: the *product* $a_{\text{now}} \cdot a_{\text{eq}}$ carries R to the power $1/3$, as required by:

$$a_0^2 = a_H^2 \cdot R^{1/3}$$

4.3 The approximate bridge

Unlike the Planck and Compton cases, the two cosmological accelerations do not scale as exact inverse powers of R . The current-epoch acceleration carries $R^{1/2}$, while the equality-epoch acceleration carries $R^{-1/6}$. The exponents are:

$$\alpha \sim 1/2, \quad \beta \sim 1/6$$

yielding $\varepsilon = \alpha - \beta \sim 1/3$, so:

$$a_0 = a_H \cdot R^{(\alpha+\beta)/2} \approx a_H \cdot R^{1/6}$$

This is a Case II (approximate) bridge. The geometric mean suppresses the parameter dependence — from a $1/2$ power (the dominant scaling of either individual acceleration) down to a $1/6$ power. The density partition R enters only weakly.

4.4 Physical interpretation

The MOND acceleration $a_0 \approx 1.2 \times 10^{-10} \text{ m/s}^2$ is nearly equal to $cH_0/(2\pi) \approx 1.1 \times 10^{-10} \text{ m/s}^2$, differing by the factor $R^{1/6} \approx 1.15$. The base scale a_H is the leading term; the $1/6$ -power residual is the scar left by imperfect cancellation in the approximate bridge.

This explains a longstanding puzzle: *why* $a_0 \approx cH_0$. It is not a coincidence. The MOND scale is the geometric-mean fixed point of two cosmological accelerations, and the geometric mean suppresses cosmological parameter dependence, yielding a nearly universal acceleration scale from epoch-dependent inputs.

4.5 Comparison with exact bridges

Property	Planck (Exact)	Compton (Exact)	MOND (Approximate)
Scale A	$\ell_{\text{grav}} = (\hbar/mc)$ $(m/m_P)^2$	$a_B = \tilde{\kappa}_e / \alpha$	$a_{\text{now}} \sim a_H \cdot R^{(1/2)}$
Scale B	$\ell_{\text{quantum}} = \hbar/(mc)$	$r_e = \tilde{\kappa}_e \cdot \alpha$	$a_{\text{eq}} \sim a_H \cdot R^{(-1/6)}$
Running parameter p	m/m_P	α_{em}	$R = \Omega_\Lambda/\Omega_m$
α exponent	1	1	$\sim 1/2$
β exponent	1	1	$\sim 1/6$
$\varepsilon = \alpha - \beta$	0	0	$\sim 1/3$
Bridge result	ℓ_P (universal)	$\tilde{\kappa}_e$ (universal for given m)	$a_0 \approx a_H \cdot R^{(1/6)}$ (soft constant)
Residual dependence	None	None	$\sim 15\%$ from density partition

5. Discussion

5.1 Extending the Padmanabhan program

Padmanabhan (2004) identified $\rho_{\text{vac}} \approx \sqrt{(\rho_{\text{UV}} \cdot \rho_{\text{IR}})}$ as a geometric-mean relationship linking the cosmological constant to the Planck and Hubble scales. That observation addressed the *value* of Λ . The present paper extends the same structural pattern to the *value* of a_0 , showing that the MOND acceleration is a geometric mean of cosmological accelerations at two epochs.

The key difference is that Padmanabhan's cosmological-constant bridge is exact (or nearly so — the observed ρ_{vac} matches the geometric mean within observational uncertainties), while the MOND bridge is explicitly approximate, with a 1/6-power residual dependence on Ω_Λ/Ω_m . The exact/approximate classification introduced in Section 2 formalizes this difference: Padmanabhan's result is Case I, the MOND result is Case II, and the residual exponent ε encodes the degree of asymmetry between the two competing scales.

Together, these results suggest that geometric-mean bridging is a recurring structural feature of cosmological scale hierarchies, not a one-off observation. The Planck length, the dark energy

density, and the MOND acceleration all sit at or near geometric-mean fixed points of competing scales — spanning quantum gravity, cosmology, and galactic dynamics.

5.2 Exact versus approximate: a diagnostic tool

The distinction between Case I and Case II bridges may serve as a diagnostic. An exact bridge ($\varepsilon = 0$) indicates that the two scales arise from a single underlying symmetry — the opposite power-law dependencies are exactly matched, suggesting a deep structural reason. An approximate bridge ($\varepsilon \neq 0$) indicates that the two scales arise from related but distinct physics, with the residual exponent encoding how far the system is from exact symmetry.

For MOND, the $1/6$ -power residual may encode information about the relationship between dark energy and matter that is not captured by the leading-order cH_0 approximation. This is a testable prediction: if the geometric-mean bridge is the correct structural explanation for a_0 , then the precise cosmological-parameter dependence of a_0 should follow the $(\Omega_\Lambda/\Omega_m)^{1/6}$ form derived in Papers 1–5, rather than alternative scalings.

5.3 Possible further instances

The geometric-mean bridge pattern may appear at other hierarchy crossings:

- **The Chandrasekhar mass** $M_{\text{Ch}} \sim m_P^3/m_p^2$ sits between quantum degeneracy pressure and gravitational collapse, with the nucleon mass as the running parameter. This is a GM-flavored combination, though it involves $3/2$ powers rather than a clean square root.
- **The Jeans length** $\lambda_J \sim c_s \sqrt{(\pi/G\rho)}$ bridges thermal pressure and gravitational collapse — the same structural pattern applied to astrophysical fluid dynamics.
- **The electroweak scale** may admit a similar decomposition if the hierarchy problem is recast in terms of competing scalings with the Planck scale.

These remain speculative but testable directions for extending the framework.

5.4 What this paper does not claim

This paper does not claim that the geometric-mean bridge is a dynamical mechanism. It is a structural observation: certain fundamental scales *can be written as* geometric means of competing scales, and this mathematical fact *explains* their parameter-insensitivity. Whether there is a deeper dynamical principle that *requires* nature to select geometric-mean fixed points — or whether this is simply a convenient mathematical decomposition — remains an open question.

6. Conclusion

Padmanabhan (2004) showed that the dark energy density sits at the geometric mean of Planck-scale and Hubble-scale energy densities. This paper extends that observation to the MOND

acceleration scale, showing that a_0 is the geometric mean of two cosmological accelerations — at the current (Λ -dominated) epoch and at matter-radiation equality.

The extension reveals a structural distinction. The Planck length and Compton wavelength are exact geometric-mean bridges: the running parameter (mass, electromagnetic coupling) cancels entirely, yielding universal constants. The MOND acceleration is an approximate bridge: the running parameter (Ω_Λ/Ω_m) is suppressed from a $\sim 1/2$ power to a $1/6$ power, but survives as a weak residual. This exact/approximate classification formalizes the difference between l_P (which depends on no physical parameter) and a_0 (which depends weakly on cosmological parameters).

The practical consequence: the 40-year "coincidence" $a_0 \approx cH_0$ is the leading-order term of a geometric-mean bridge. The $\sim 15\%$ deviation from $cH_0/(2\pi)$ is not noise — it is the $(\Omega_\Lambda/\Omega_m)^{1/6}$ correction predicted by the approximate bridge structure, encoding the asymmetry between the two epoch-dependent accelerations whose geometric mean produces a_0 .

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